NeuroLens

Watch a brain watch your video.

PREDICTED BRAIN RESPONSE TO VIDEO, SECOND BY SECOND

CS5224 CLOUD COMPUTING · FINAL PRESENTATION · DYLAN CHAN · XIN WEI CHONG · KELVIN LOW · JOSHUA LUM · BRANDON TAN

$10k+

to test one ad on a brain scanner

fMRI study	**$10 – 15k**	2–4 weeks
NeuroLens	**$1.00**	≈ 7 min

Ten cuts by lunch.
No way to judge them.

Generative tools made production cheap. The feed made every second count.



What already exists: measured, filmed or guessed.

Brain scans (fMRI, EEG) research labs, iMotions suites	$10,000 – $15,000	2–4 weeks
Facial coding Realeyes, iMotions webcam studies	$850 – $4,000	3–5 days
AI attention maps Neurons AI, Attention Insight	$500 – $1,500 / mo	sub-minute
NeuroLens predicted brain response	$1.00	≈ 7 min

OUR DIFFERENCE

A brain response, not a gaze or a face

Per second, with a no-sound pass

Compare any two cuts for $1.00

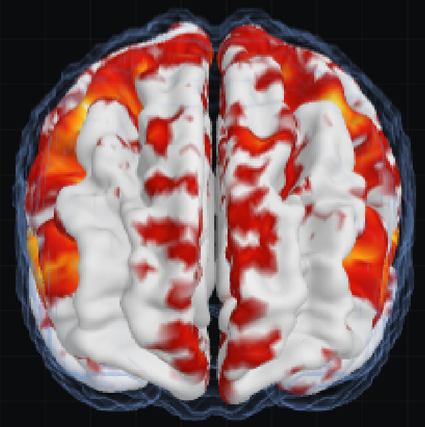
Under the hood: TRIBE v2.

Frames
V-JEPA 2 watches

Sound
Wav2Vec-BERT listens

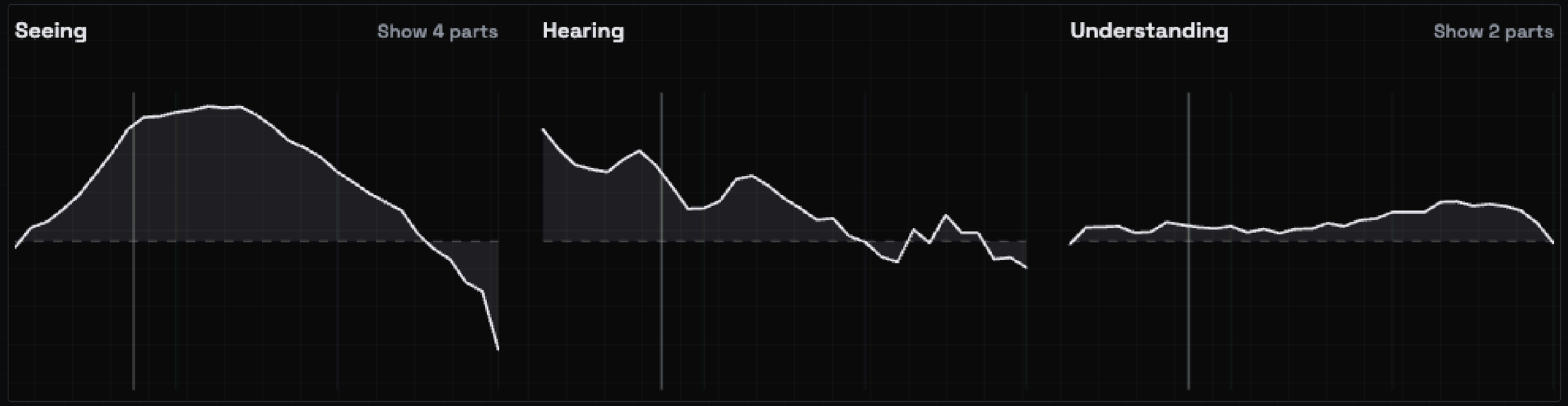
Words
WhisperX + LLaMA 3.2 read

TRIBE v2
1,000+ hours of fMRI
from 720 people



20,484 cortex points, every second

Seven signals, three questions.



SEEING

faces · bodies · scenes · visual

HEARING

sound

UNDERSTANDING

speech · social

BELOW TYPICAL

0 = TYPICAL FILM & TV VIEWING

ABOVE TYPICAL

Pay as you go. Credit, not cards.

$1.00

per started 30 s of video

NEW USERS: $10 CREDIT = TEN 30 S VIDEOS

WHY PREPAID CREDIT: CARD FEE AS A SHARE OF THE CHARGE

A $1.00 video, paid by card

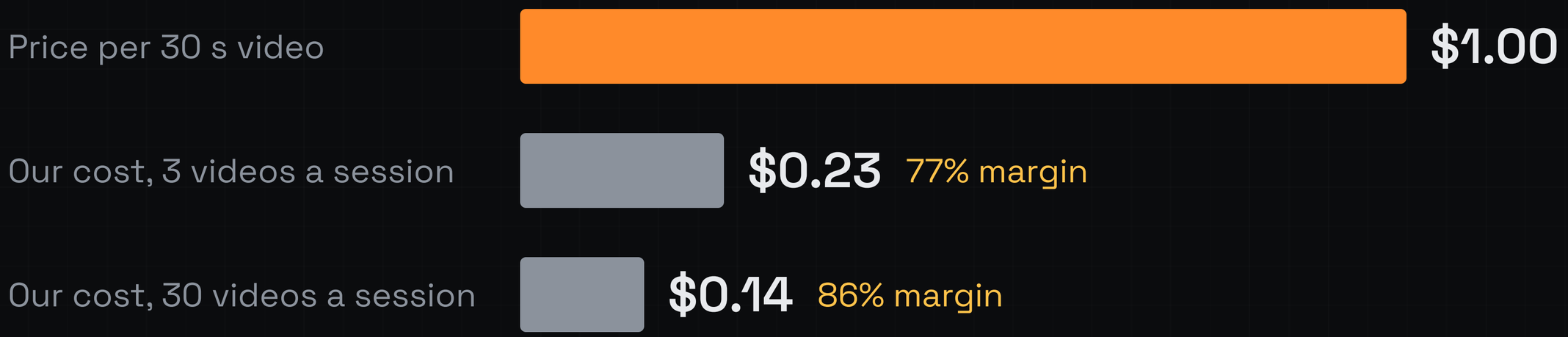
≈ 33%

A $20 credit pack

≈ 4%

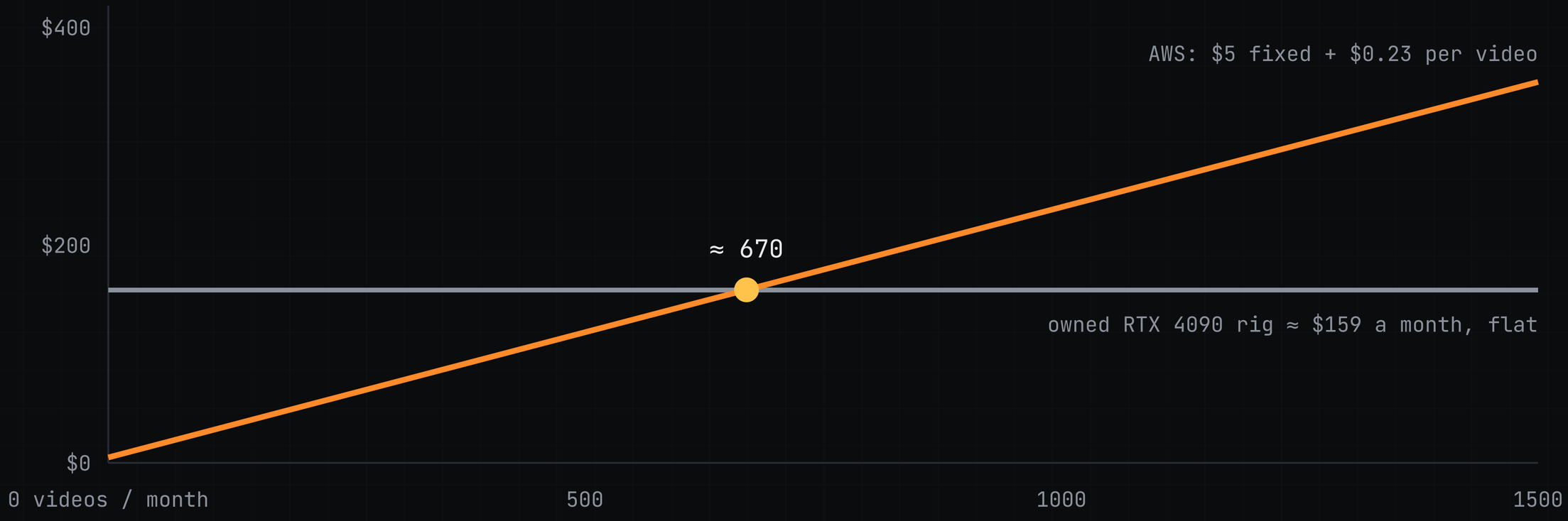
COURSE VERSION: AN ADMINISTRATOR GRANTS CREDIT · STRIPE CHECKOUT BELONGS TO THE PRODUCTION DESIGN

What one analysis costs us.

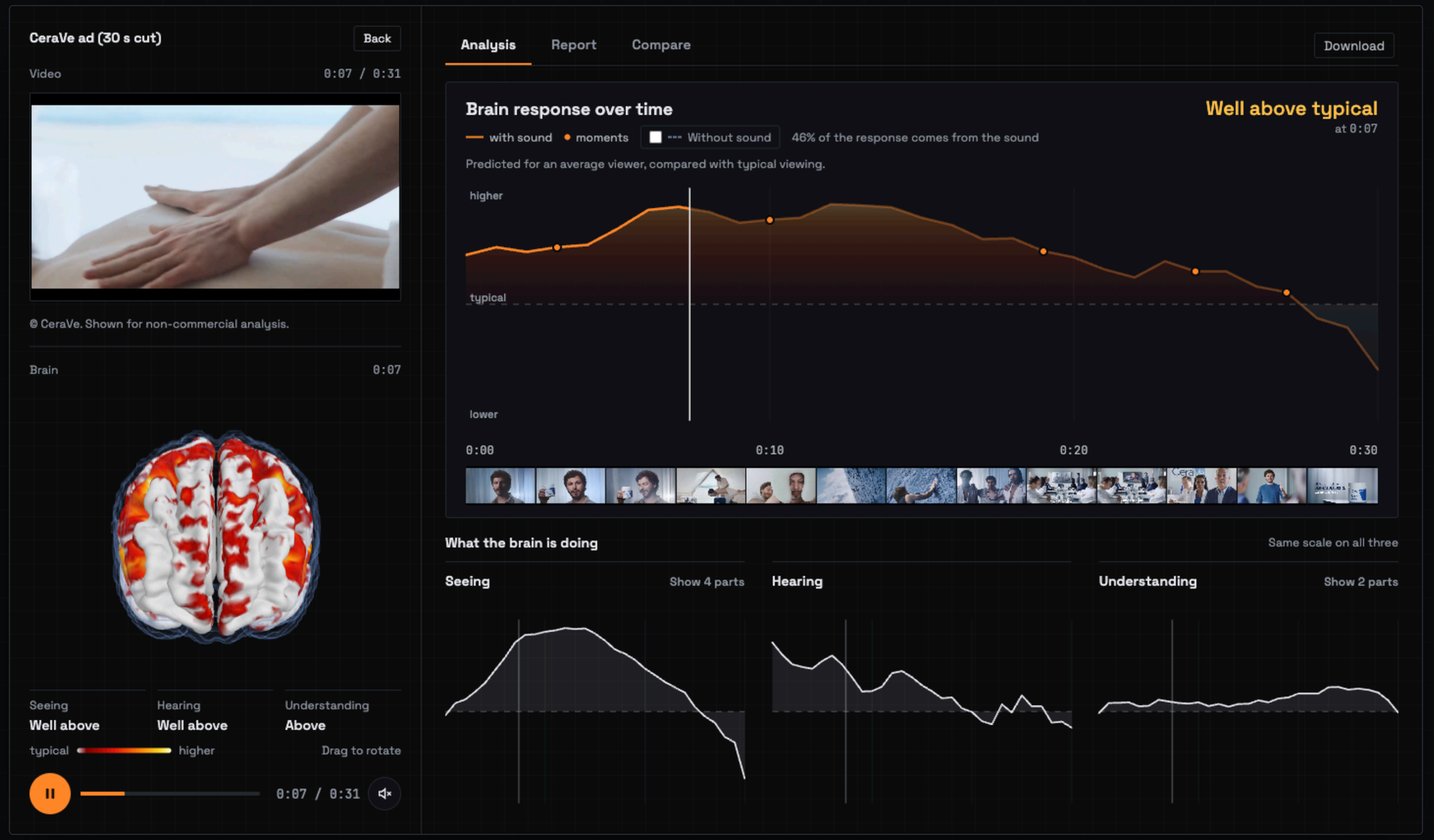


GPU 393 S AT $1.22/H (L4) · START-UP AND IDLE SHARED ACROSS A SESSION · FIXED COST $5 A MONTH

Cloud is cheaper below about 670 videos a month.



LIVE Live demo: sign in, upload, explore



1. Upload: price first
2. Peak, dip, no-sound toggle
3. Seeing · hearing · understanding
4. AI report, compare, bad file

CLICK THE PICTURE TO OPEN THE LIVE SITE · DF6GIRQYYKYFA.CLOUDFRONT.NET

How do we know it works?

THE MODEL: VALIDATED BY META

720

people scanned; accurate on new videos, tasks and people. Not re-tested by us.

OUR PIPELINE

0.005

largest difference, RTX 4090 vs AWS, across 11 series

77/80

refund tests passed

OUR USERS

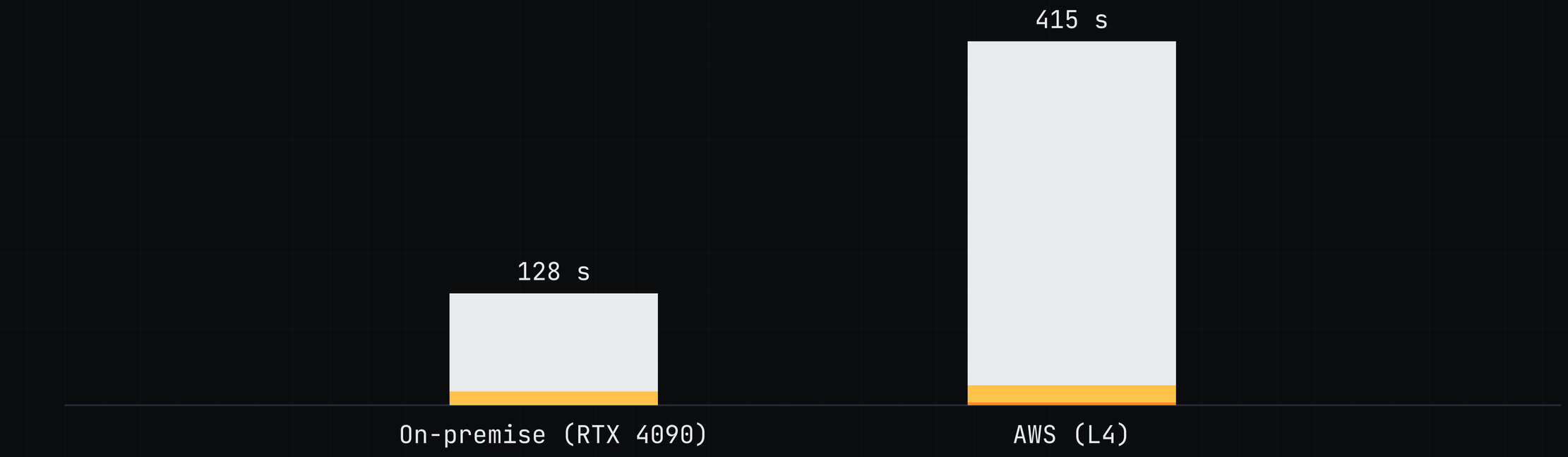
[TBD]

SUS score and survey: see Usability

Not validated: that a higher response means more views or sales.

Where does the time go in one job?

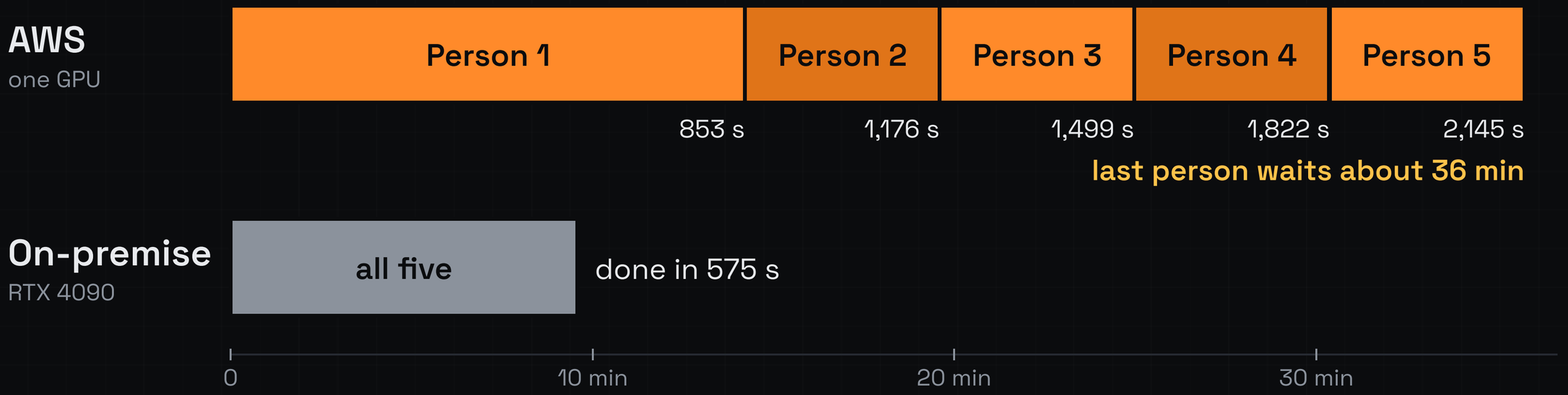
Same code, same 30 s clip, on our own RTX 4090 and on AWS.



GPU TIME PER VIDEO-SECOND: 3.71 S (RTX 4090) VS 10.7 S (A10G) · FIRST VIDEO AFTER 7+ MIN IDLE: 779 S ON AWS

Five people at once: they wait in line.

Five uploads at the same moment. One GPU runs one job at a time.



NOTHING WAS DROPPED. THE AWS RUN GOT AN A10G, AND ITS FIRST JOB APPEARS TO HAVE RESTARTED ONCE.

Can people use it?

Booth visitors tried a sample or uploaded a clip, then answered the 10-question SUS.

[TBD] / 100

SUS SCORE · 68 IS AVERAGE · 80 IS VERY GOOD

[X]% SAW SOMETHING THEY HAD NOT NOTICED
[Y]% WOULD PAY $1.00 FOR A 30 S VIDEO

What it claims, and what it doesn’t.

WHAT IT IS

- An average viewer’s brain response
- One scale for every video

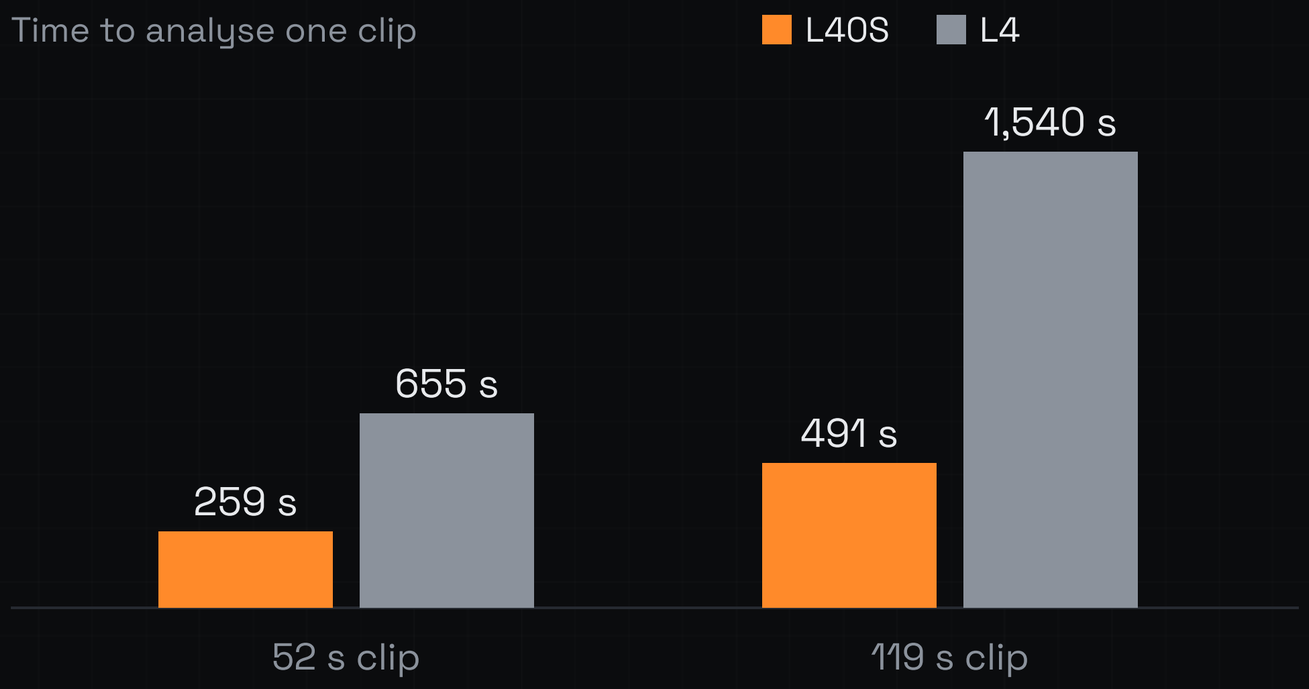
WHAT IT ISN'T

- A prediction of views or sales
- Instant: first video ≈ 6 min

ETHICS

- Non-commercial licence (CC BY-NC)
- No one scanned or profiled
- Anonymous, consented survey

Where next: scaling up.



26 / 26

Spot launches failed for lack of capacity

19.9 GB

peak GPU memory: a 24 GB card fits

An EC2 GPU pool: $0.10 – $0.20 a video, about $70 more a month. It pays off above about 1,000 videos a month. Estimates, not built.

A $10,000 brain study, in 7 minutes for $1.

$1.00
per 30 s video

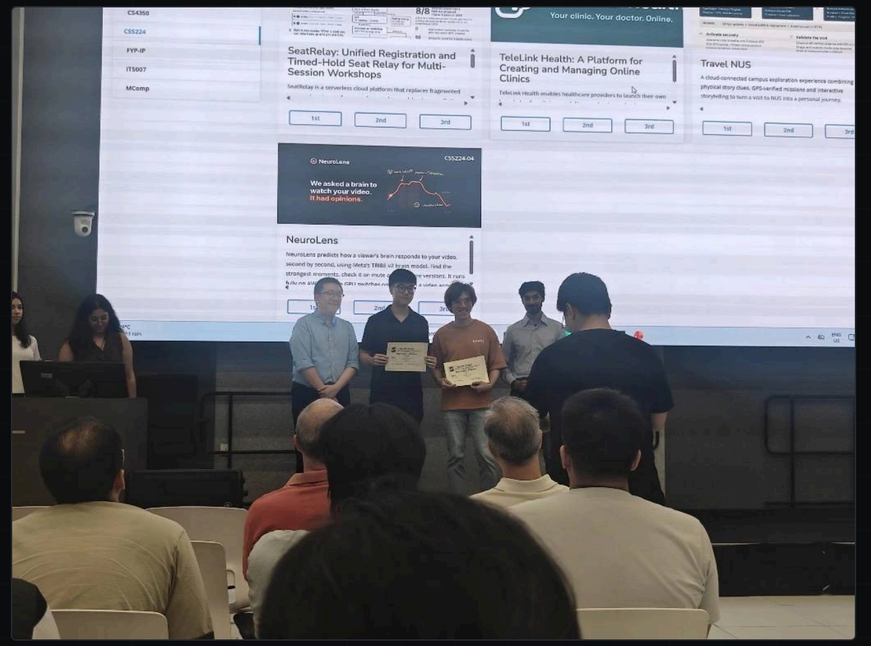
≈ 7 min
to a result, GPU warm

670
videos a month: break-even vs owning

≈ 6 min
first video after a quiet spell

NEXT

A commercial licence · a GPU pool that grows with the queue · fixes from our usability study



Thanks for supporting us at STePS!

Thank you.
Questions?

df6girqyykyfa.cloudfront.net

CODE · GITHUB.COM/KELVINLOWHM86/NEUROLENS